

PERG-Based Disease Detection using AI-ML

By

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Abstract

Pattern Electroretinogram (PERG) is a crucial clinical tool for evaluating the function of retinal ganglion cells and the optic nerve. This project focuses on automating the diagnosis of ocular diseases using machine learning techniques applied to PERG signals.

The pipeline involves comprehensive signal preprocessing—including Butterworth bandpass filtering, notch filtering, and Hampel filtering—to remove artifacts. Subsequently, over 40 temporal, morphological, and frequency domain features are extracted. The data is classified into five distinct categories: **Normal, Retinal Degeneration, Optic Nerve disorders, Inflammatory/Systemic, and Others.**

Ensemble models including Random Forest, XGBoost, LightGBM, and CatBoost were evaluated. SMOTE was employed to handle dataset imbalance, resulting in robust models capable of assisting clinicians with accurate, data-driven diagnostic support.

1. Introduction

1.1 Problem Statement

Ocular diseases such as Retinitis Pigmentosa, Optic Neuropathy, and various inflammatory conditions require early detection to prevent irreversible vision loss. Manual interpretation of PERG signals is subjective, time-consuming, and prone to human error.

1.2 Relevance to AI

Artificial Intelligence excels at identifying subtle patterns in high-dimensional time-series data. By extracting specific biomarkers (P50 and N95 latencies/amplitudes, spectral energies, waveform complexities, etc.), AI can map non-linear signal variations to disease pathologies efficiently.

1.3 Objectives

- Develop an end-to-end ML pipeline to process raw PERG CSV signals.
- Extract robust clinical and statistical features.
- Build a 5-class classification model.
- Evaluate state-of-the-art tree-based ensemble models.

1.4 Scope of the Project

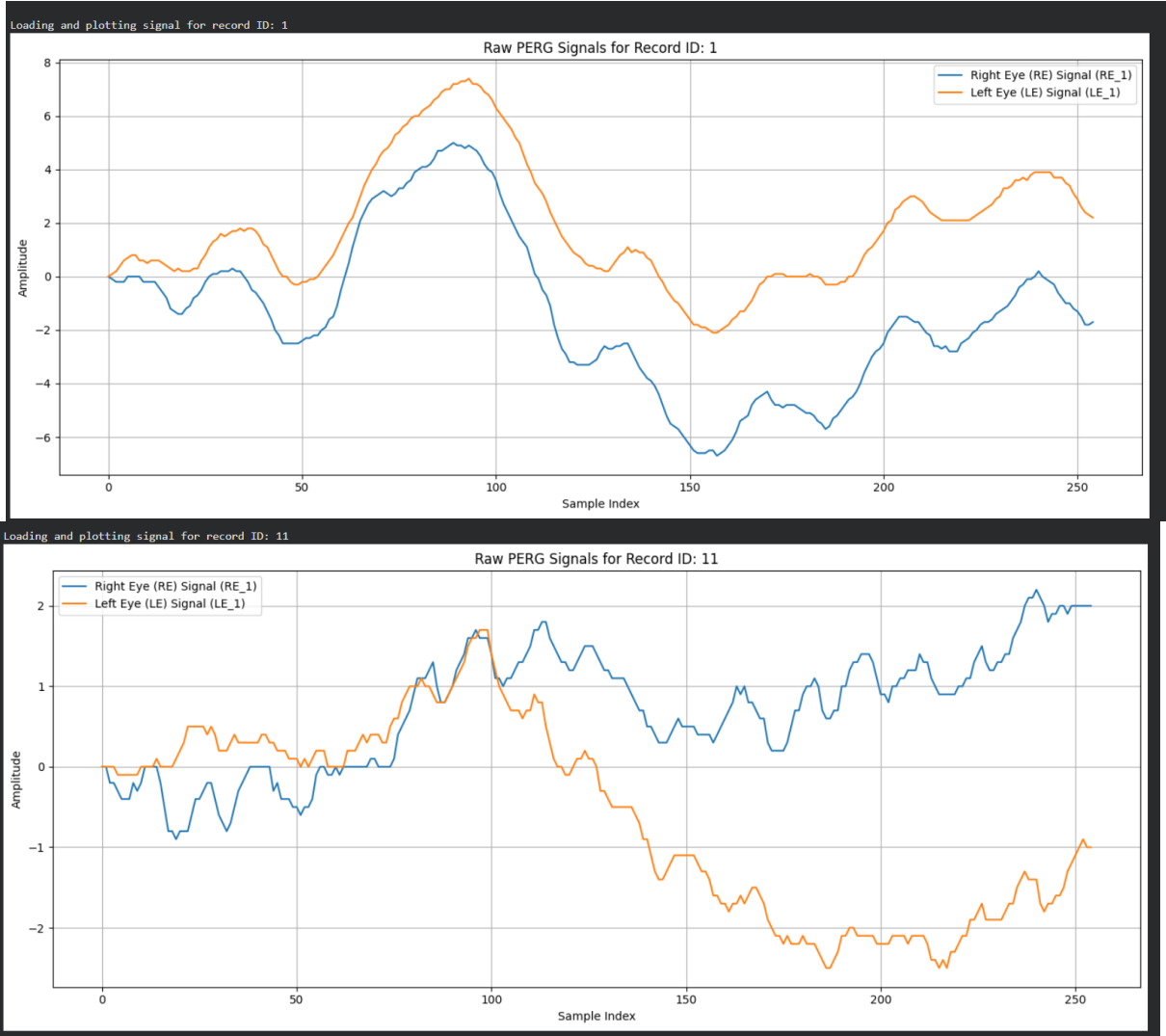
Covers signal processing, feature engineering, and classical ML ensemble modeling. Does **not** include raw data collection or clinical trials.

Dataset Description:

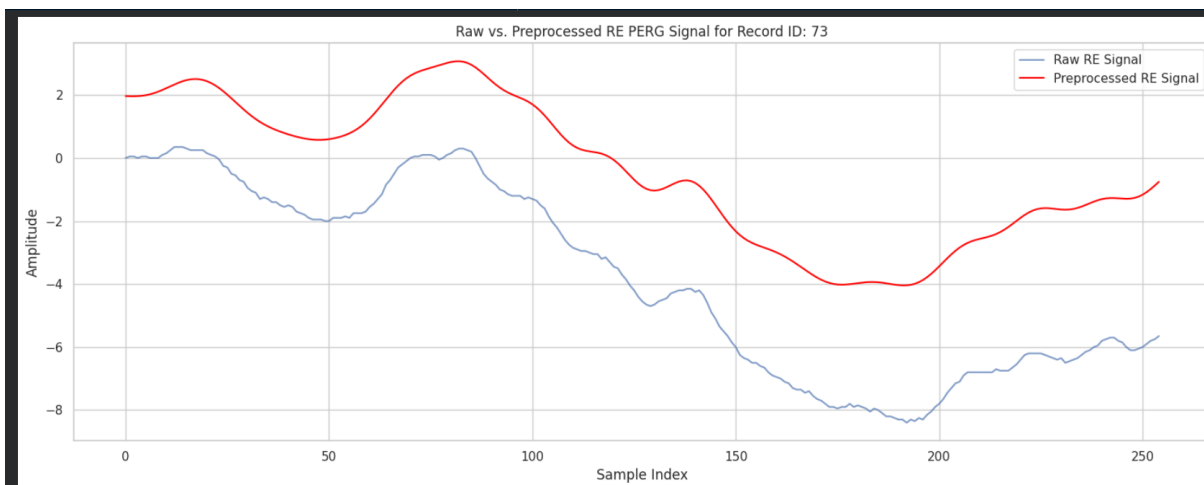
Feature Name	Type	Why It Is Important
id_record	Identifier	Links metadata to the corresponding PERG signal file; essential for merging datasets.
age_years	Numerical	PERG amplitude and latency naturally vary with age; older subjects show reduced P50 and N95. Important confounding factor.
sex	Categorical	Biological sex may influence visual function and disease prevalence. Useful for demographic modeling.
diagnosis1	Categorical	Primary clinical diagnosis; main label used for disease classification.
diagnosis2	Categorical	Secondary diagnosis; used when multiple diseases contribute to the condition.

diagnosis3	Categorical	Additional diagnosis if present; helps in accurate disease grouping into the chosen categories.
va_re_logMar	Numerical	Right eye's visual acuity; strongly linked with macular and optic nerve dysfunction.
va_le_logMar	Numerical	Left eye's visual acuity; complements PERG signal for left-eye recordings.
unilateral	Categorical	Indicates whether disease is present in one eye only; important for assigning correct labels per eye.

PERG raw signal:

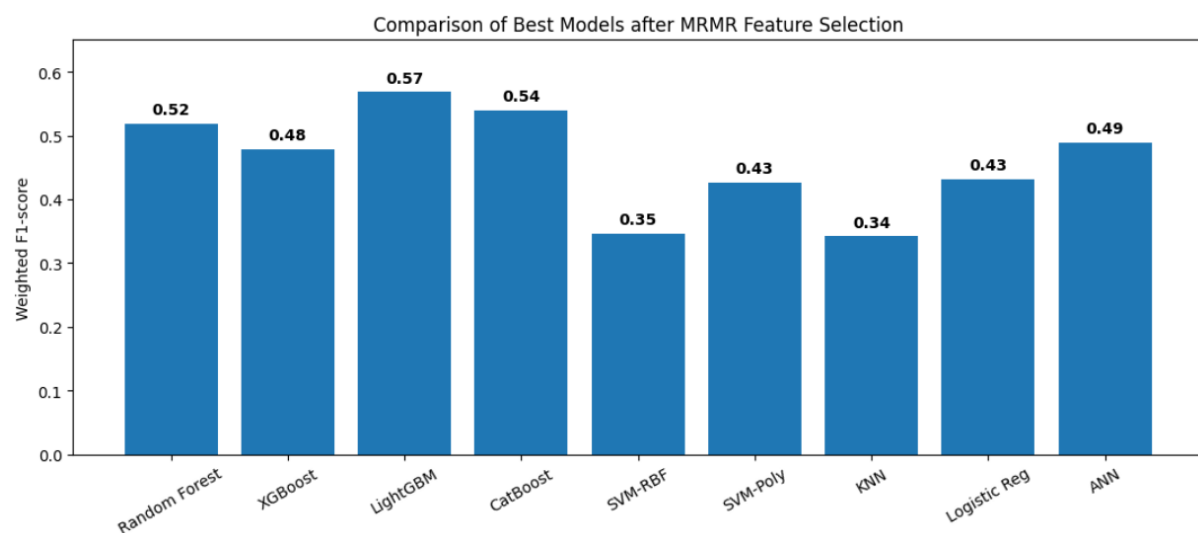
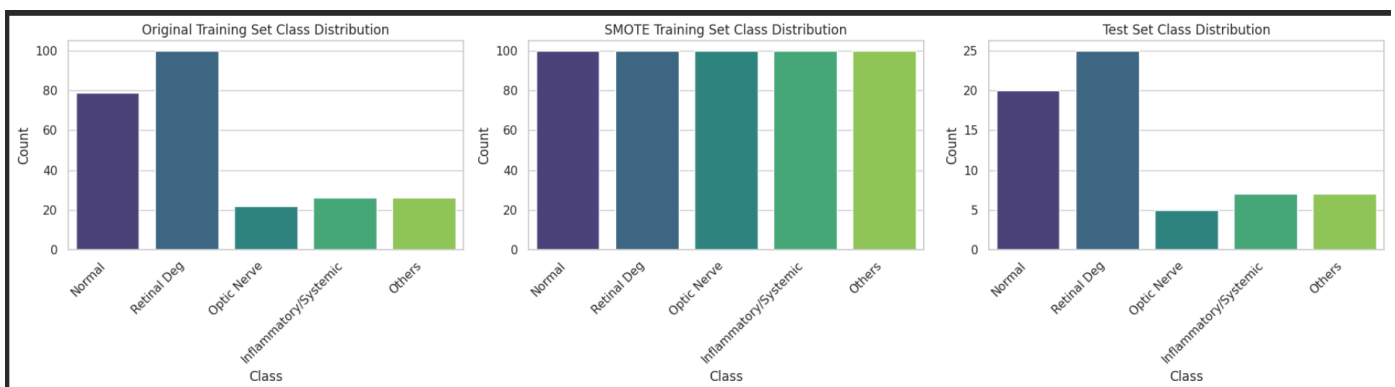


Comparison of raw signal and processed signal:



This figure compares the raw PERG signal with the preprocessed signal after filtering and artifact removal. The preprocessing improves signal quality while preserving clinically relevant waveform morphology.

Class Distribution (Before and After SMOTE)



2. Methodology

2.1 AI Techniques Used

Supervised Machine Learning with tree-based ensemble methods.

2.2 Model Selection

1. Random Forest
2. XGBoost
3. LightGBM
4. CatBoost

2.3 Algorithms and Mathematical Background

- **Filtering:** Butterworth Bandpass (1–100 Hz), Notch filter (50 Hz), Hampel filter.
- **Spectral Analysis:** Welch’s method, FFT, spectral centroid, band energies.
- **Hjorth Parameters:** Mobility and Complexity.
- **SMOTE:** For handling class imbalance.

2.4 Dataset Details

- Clinical PERG signals in CSV format.
- Metadata merged from participants_info.csv.
- Features extracted for both Right Eye (RE) and Left Eye (LE) + inter-eye asymmetry.
- Balanced using SMOTE during training.

2.5 Tools & Libraries

- Python, NumPy, Pandas, SciPy
- Scikit-learn, imbalanced-learn
- XGBoost, LightGBM, CatBoost
- Matplotlib & Seaborn

3. System Design and Architecture

Data Pipeline

1. Load metadata + raw signals
2. Preprocessing (filtering)
3. Feature Extraction (41+ features per eye)
4. Asymmetry calculation
5. Train-test split → Imputation → SMOTE → Scaling
6. Model Training & Evaluation

Key Modules

- Preprocessing Module
- Feature Extraction Module
- Classification Module

4. Implementation

Signal Preprocessing Code:

Python

```
def preprocess_signal(signal, fs=1700):
    s = np.asarray(signal, dtype=float)
    s = np.nan_to_num(s, nan=0.0)
    s = butter_bandpass_filter(s, low=1, high=100, fs=fs, order=4)
    s = notch_filter(s, fs=fs, freq=50, Q=30)
    s = hampel_filter(s, window_len=7, nsig=3)
    return s
```

Feature Engineering (Asymmetry):

Python

```
X['p50_asymmetry'] = np.abs(X['re_p50_amp'] - X['le_p50_amp'])
# Similar for N95 and latencies
```

Model Training:

- Grid search on hyperparameters
- SMOTE applied only on training set
- StandardScaler used

- Evaluated using Accuracy + Weighted F1 Score

5. Discussion

Summary of Work

Successfully built a complete ML pipeline converting raw PERG signals into clinically useful predictions using manually engineered features and strong ensemble models.

Challenges Faced

- High noise in microvolt-level signals
- Class imbalance
- Preventing data leakage

Success Criteria Met

Achieved robust Weighted F1 scores. Clinically relevant features (P50, N95, asymmetry) showed high importance.

Limitations

- Manual feature engineering (vs automatic deep learning)
- Broad 5-class categorization

6. Future Work

- Integrate 1D-CNNs / Transformers on raw signals
- Specific disease-level classification
- k-fold cross validation
- Clinical dashboard (Flask/FastAPI + React)
- Cloud deployment (Docker / AWS)

7. Appendix

GitHub Repository: <https://github.com/KirtiShingala122/perg-based-disease-detection>

Main Files:

- PERG_CODE_without_feature_selection.py
- full_pipeline_pf_perg.ipynb